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Rainbow Connection of Random Regular Graphs

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Abstract

An edge colored graph G is rainbow edge connected if any two vertices are connected by a path whose edges have distinct colors. The rainbow connection of a connected graph G, denoted by $rc(G)$, is the smallest number of colors that are needed in order to make G rainbow connected.

In this work we study the rainbow connection of the random r-regular graph $G = G(n, r)$ of order n, where $r \geq 4$ is a constant. We prove that with probability tending to one as n goes to infinity the rainbow connection of G satisfies $rc(G) = O(\log n)$, which is best possible up to a hidden constant.

1 Introduction

Connectivity is a fundamental graph theoretic property. Recently, the concept of rainbow connection was introduced by Chartrand, Johns, McKeon and Zhang in [8]. We say that a set of edges is *rainbow colored* if its every member has a distinct color. An edge colored graph G is *rainbow edge connected* if any two vertices are connected by a rainbow colored path. Furthermore, the *rainbow connection* $rc(G)$ of a connected graph G is the smallest number of colors that are needed in order to make G rainbow edge connected.

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Notice, that by definition a rainbow edge connected graph is also connected. Moreover, any connected graph has a trivial edge coloring that makes it rainbow edge connected, since one may color the edges of a given spanning tree with distinct colors. Other basic facts established in [8] are that $rc(G) = 1$ if and only if G is a clique and $rc(G) = |V(G)| - 1$ if and only if G is a tree. Besides its theoretical interest, rainbow connection is also of interest in applied settings, such as securing sensitive information transfer and networking (see, e.g., [6, 16]). For instance, consider the following setting in networking [6]: we want to route messages in a cellular network such that each link on the route between two vertices is assigned with a distinct channel. Then, the minimum number of channels to use is equal to the rainbow connection of the underlying network.

Caro, Lev, Roditty, Tuza and Yuster [5] prove that for a connected graph G with n vertices and minimum degree δ, the rainbow connection satisfies $rc(G) \leq \frac{\log \delta}{\delta} n(1 + f(\delta))$, where $f(\delta)$ tends to zero as δ increases. The following simpler bound was also proved in [5], $rc(G) \leq n\frac{4\log n+3}{\delta}$. Krivelevich and Yuster [15] removed the logarithmic factor from the upper bound in [5]. Specifically they proved that $rc(G) \leq \frac{20n}{\delta}$. Chandran, Das, Rajendraprasad and Varma [7] improved this upper bound to $\frac{3n}{\delta+1} + 3$, which is close to best possible.

As pointed out in [5] the random graph setting poses several intriguing questions. Specifically, let $G = G(n, p)$ denote the binomial random graph on n vertices with edge probability p. Caro, Lev, Roditty, Tuza and Yuster [5] proved that $p = \sqrt{\log n/n}$ is the sharp threshold for the property $rc(G) \leq 2$. This was sharpened to a hitting time result by Heckel and Riordan [12]. He and Liang [11] studied further the rainbow connection of random graphs. Specifically, they obtain a threshold for the property $rc(G) \leq d$ where d is constant. Frieze and Tsourakakis [10] studied the rainbow connection of $G = G(n, p)$ at the connectivity threshold $p = \frac{\log n+\omega}{n}$ where $\omega \to \infty$ and $\omega = o(\log n)$. They showed that w.h.p.[1] $rc(G)$ is asymptotically equal to $\max\{diam(G), Z_1(G)\}$, where Z_1 is the number of vertices of degree one.

For further results and references we refer the interested reader to the recent survey of Li, Shi and Sun [16].

In this paper we study the rainbow connection of the random r-regular graph $G(n, r)$ of order n, where $r \geq 4$ is a constant and $n \to \infty$. It was shown in Basavaraju, Chandran, Rajendraprasad, and Ramaswamy [2] that for any bridgeless graph G, $rc(G) \leq \rho(\rho + 2)$, where ρ is the radius of $G = (V, E)$, i.e., $\min_{x\in V} \max_{y\in V} dist(x, y)$. Since the radius of $G(n, r)$ is $O(\log n)$ w.h.p., we see that [2] implies that $rc(G(n, r)) = O(\log^2 n)$ w.h.p.The following theorem gives an improvement on this for $r \geq 4$.

[1] An event $\mathcal{E}_n$ occurs *with high probability*, or w.h.p. for brevity, if $\lim_{n\to\infty} \Pr(\mathcal{E}_n) = 1$.

Theorem 1 *Let $r \geq 4$ be a constant. Then, w.h.p. $rc(G(n,r)) = O(\log n)$.*

The rainbow connection of any graph G is at least as large as its diameter. The diameter of $G(n,r)$ is w.h.p. asymptotically $\log_{r-1} n$ and so the above theorem is best possible, up to a (hidden) constant factor.

We conjecture that Theorem 1 can be extended to include $r = 3$. Unfortunately, the approach taken in this paper does not seem to work in this case.

2 Proof of Theorem 1

2.1 Outline of strategy

Let $G = G(n,r)$, $r \geq 4$. Define

$$k_r = \log_{r-1}(K_1 \log n), \tag{1}$$

where K_1 will be a sufficiently large absolute constant. Recall that the *distance between two vertices* in G is the number of edges in a shortest path connecting them and the *distance between two edges* in G is the number of vertices in a shortest path between them. (Hence, both adjacent vertices and incident edges have distance 1.)

For each vertex x let T_x be the subgraph of G induced by the vertices within distance k_r of x. We will see (due to Lemma 5) that w.h.p., T_x is a tree for most x and that for all x, T_x contains at most one cycle. We say that x is *tree-like* if T_x is a tree. In which case we denote by L_x the leaves of T_x. Moreover, if $u \in L_x$, then we denote the path from u to x by $P(u,x)$.

We will randomly color G in such a way that the edges of every path $P(u,x)$ is rainbow colored for all x. This is how we do it. We order the edges of G in some arbitrary manner as $e_1, e_2, \ldots, e_m$, where $m = rn/2$. There will be a set of $q = \lceil K_1^2 r \log n \rceil$ colors available. Then, in the order $i = 1, 2, \ldots, m$ we randomly color e_i. We choose this color uniformly from the set of colors not used by those $e_j, j < i$ which are within distance k_r of e_i. Note that the number of edges within distance k_r of e_i is at most

$$2\left((r-1) + (r-1)^2 + \cdots + (r-1)^{\lfloor k_r \rfloor - 1}\right) \leq (r-1)^{k_r} = K_1 \log n. \tag{2}$$

So for K_1 sufficiently large we always have many colors that can be used for e_i. Clearly, in such a coloring, the edges of a path $P(u,x)$ are rainbow colored.

Now consider a fixed pair of tree-like vertices x, y. We will show (using Corollary 4) that one can find a partial 1-1 mapping $f = f_{x,y}$ between L_x and L_y such that if $u \in L_x$